Optimising Sampling Strategy for Accurate Kiwifruit Yield Estimation

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Introduction

Accurate pre-harvest yield estimation is critical to Zespri's operations across various departments. Current sampling strategy relies on Digital Orchard Scanning (DOS) method, which uses high-resolution cameras mounted on all-purposed vehicles to scan the full width of orchard rows (see Figure 1). This sampling strategy is resource-intensive, costly and lack of scalability.

Objective

To evaluate alternative methods for cost-effective sampling solutions for expanding Zespri's sampling capacity.

Figure 1. KWKIWI operation team with Digital Orchard Scanning equipment

Data

Raw data contains geospatial locations of row boundaries and kiwifruit points. 4816 sampled rows with over 100 million kiwifruit points are successfully extracted. Data preprocessing follows the steps below (Figure 2.)

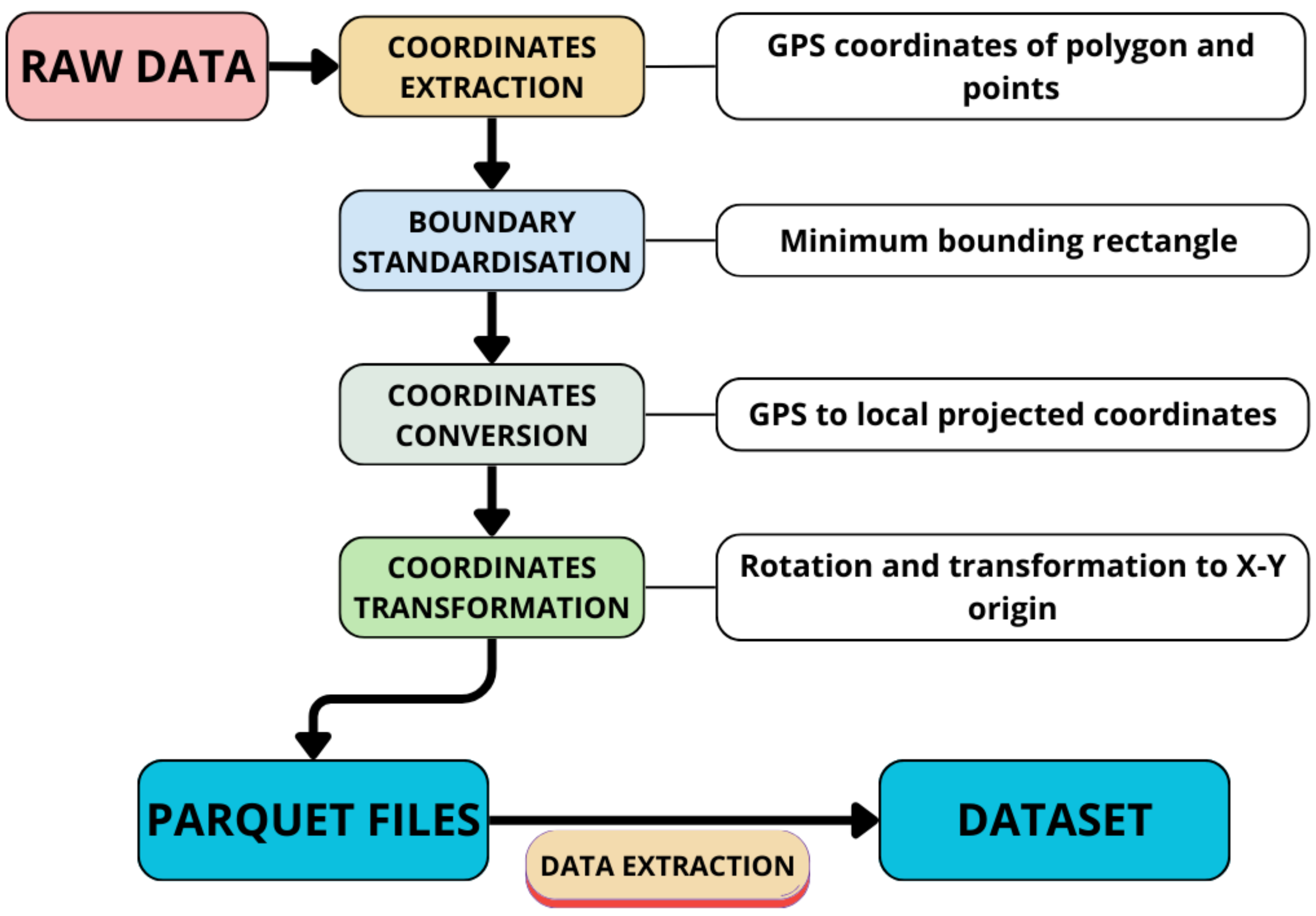

Figure 2. Data preprocessing workflow

Methods

- **Quadrat sampling:** Random quadrat placement is applied as shown in Figure 3. Varied quadrat shape, size and count configurations are evaluated.
- **Row-centre-only sampling:** A fix wide strip along the row centre is used to simulate reduced resource requirement. One sample T-test is conducted to identify the significance of bias.
- **Machine Learning integration:** Sampling data and geospatial features are combined to train the models (Linear Regression & Random Forest Regression).

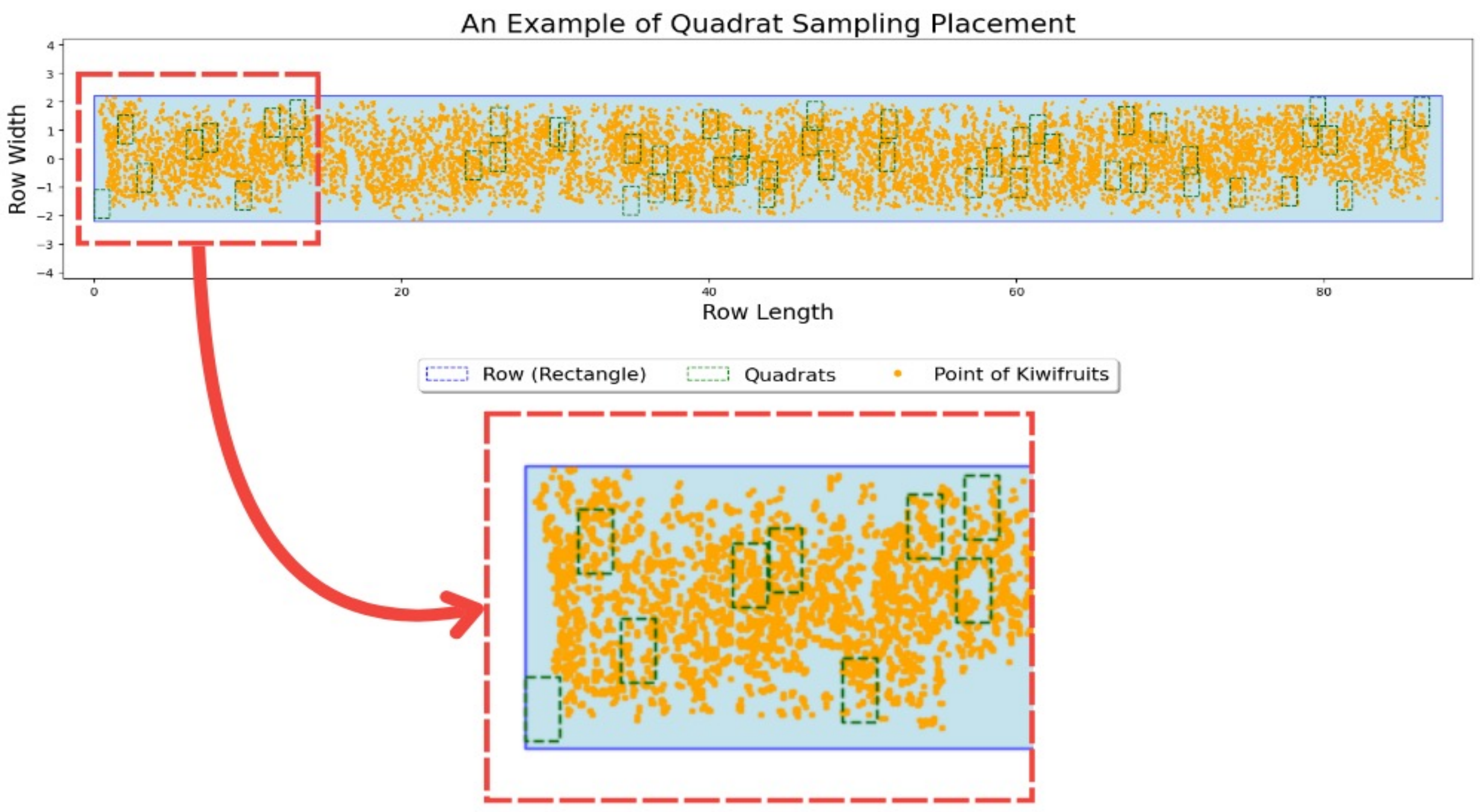

Figure 3. An example of quadrat sampling placement

Results

- **Quadrat Sampling:** Larger quadrats tend to have lower variance. Optimal quadrat configuration: 3x3 m Square quadrats with 10 samples.
- **Row-centre-only sampling:** Significant bias – overestimates density by an average of 45.64%. High variability against full-width sampling.
- **Machine Learning integration:** High estimation accuracy (see Table 1). Linear Regression overperforms Random Forest Regression.

Metric	Linear Regression	Random Forest Regression
Mean Absolute Error (MAE)	1.238	1.572
Root Mean Squared Error (RMSE)	1.768	2.096
R^2 Score	0.974	0.964
Estimation Accuracy (±10%)	94.18%	87.23%
Estimation Accuracy (±5%)	78.82%	58.15%

Table 1. Model performance summary

Conclusion

A hybrid strategy—using quadrat sampling for calibration, row-centre sampling for routine data collection, and machine learning for yield prediction—emerges as a promising solution. This approach balances precision with operational feasibility, supporting innovation and sustainability in precision agriculture.

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